

Fig. 3. (*a*) Spectrum of main discharge in nitrogen; and (*b*) spectrum of afterglow, both at a pressure of 43 mm.

Full details and a discussion of these and other results at present under investigation will be published elsewhere. This work has been possible because of the generous loan of the radar set by the Defense Research Board of Canada, and the awards of a bursary to J. R. Kenney by the National Research Council of Canada and of a scholarship to W. C. Nixon by the Research Council of Ontario.

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Diamagnetic Susceptibility and Anisotropy of Ice

The diamagnetic susceptibility of ice has been found[1] to have the mean value at 0° C. of $-12{\cdot}65 \times 10^{-6}$ c.g.s. e.m.u./mole. This is very similar to the value of the mean susceptibility of water of crystallization in organic molecules. For example, succinic acid $(CH_2.COOH)_2$, and oxalic acid hydrate $(H_2O.COOH)_2$, have mean susceptibilities of $-54{\cdot}45$ [2] and $-56{\cdot}09$ [3] ($\times 10^{-6}$), respectively. The mean susceptibility of the CH_2 group is known to be $-11{\cdot}8$, and application of Pascal's additive rule would give, by simple substitution, the value of $-12{\cdot}6$ for H_2O.

No measurements appear to have been made of the diamagnetic anisotropy of single ice crystals. The strength of ice differs considerably parallel and perpendicular to the principal axis *c*. This is because, for cleavage or slipping parallel to the *c* axis, two bonds per unit cell must be broken, as compared with one for perpendicular directions. The only bonds in ice are long hydrogen bonds; if, therefore, the diamagnetic susceptibility of ice or of water of crystallization were a directive property of the hydrogen bond, one might expect a considerable anisotropy. If it were a property only, or mainly, of the water molecules themselves, very little anisotropy would be expected.

In fact, the diamagnetic anisotropy is almost negligibly small; so small that it is difficult to measure. Blocks of ice were cut from a sheet grown naturally in cold weather on the surface of a dish of London tapwater; these were from 13 mm. × 5 mm. × 3 mm. upwards in size, and X-rays showed that the *c* axis was parallel to the smallest dimension (that is, perpendicular to the surface of the ice sheet), and that they were single crystals. The magnet pole-pieces were packed round with solid carbon dioxide in order to attain a temperature at which the measurements could be made; if the preliminary experiments had shown that an anisotropy of some considerable relative magnitude existed, then a more permanent and adjustable temperature control would have been set up. The temperature actually varied from about $-5°$ to $-15°$ during the course of the readings. The quartz fibres used could not be very fine, because of the experimental difficulty of manipulating the large ice crystals; that finally used was one which required an angle of twist (Krishnan and Banerjee's displacement method[4]) of 3,200° for a benzil crystal of mass 0·0154 gm. and anisotropy $45{\cdot}6 \times 10^{-6}$. The ice crystal, the mass of which was determined sufficiently exactly from its volume and density, was fastened to the lower end of the calibrated quartz fibre, with the *c* axis horizontal, by means of a very small stirrup of filter paper, the anisotropy of which was also measured. No other means of suspension was found to be satisfactory, and, indeed, this was not at all ideal, for the anisotropy of the stirrup was comparable with that of the ice, and unless the paper was frozen on to the ice so that their anisotropies added up, the composite effect was that the crystal would not rest with its *c* axis either axial or equatorial in the magnetic field. In fact, it behaved very much like an isotropic crystal of lithium fluoride of the same dimensions which was compared with it under similar conditions.

The largest positive result observed, for a crystal which did set definitely with its *c* axis equatorial, was $\chi_a - \chi_c = 0{\cdot}06 \times 10^{-6}$, giving the separate values:

$$\chi_c = -12{\cdot}68 \times 10^{-6} \text{ c.g.s. e.m.u./mole.}$$
$$\chi_a = -12{\cdot}62 \times 10^{-6} \text{ c.g.s. e.m.u./mole.}$$

It may be mentioned that the difference of the refractive indices in these two directions in ice-crystals is also extremely small[5].

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[1] Cabrera, B., and Fahlenbrach, H., *Naturwiss.*, **22**, 417 (1934).
[2] Gray, F. W., and Birse, W. M., *J. Chem. Soc.*, **105**, 2707 (1914).
[3] Lonsdale, K., *J. Chem. Soc.*, 365 (1938).
[4] Krishnan, K. S., and Banerjee, S., *Phil. Trans. Roy. Soc.*, A, **234**, 265 (1935).
[5] Merwin, H. E., "Intern. Crit. Tables", **7**, 17 (1930).

Positions of Three Discrete Sources of Galactic Radio-Frequency Radiation

In a recent communication[1] an account was given of the discovery of a number of discrete sources of galactic radio-frequency radiation. Accurate measurements of the position of three of these sources have since been made from sites on the east and west coasts of New Zealand and on the east coast of Australia. The technique employed was to observe the sources at rising or setting, with an aerial on a high cliff overlooking the sea. These observations, when corrected for atmospheric refraction, allow the path of a source above the horizon to be plotted, and the time of its rising and setting—and hence its celestial co-ordinates—to be determined.

It is found that all three sources correspond within limits of experimental error to positions of certain nebulous objects. The positions of the sources

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Source	Position (Epoch 1948)		Possible associated visible object		
	Right ascension	Declination	Object	Spectrum	Remarks
Taurus A	5h 31m 00s ± 30s	+ 22° 01′ ± 7′	N.G.C. 1952 (Messier 1)	Continuous. Weak emission lines of H, He, forbidden lines of N, O and Si	The Crab nebula, expanding shell of an old super-nova
Virgo A	12h 28m 06s ± 37s	+ 12° 41′ ± 10′	N.G.C. 4486 (Messier 87)	Continuous	Spherical nebula—unresolved
Centaurus A	13h 22m 20s ± 60s	− 42° 37′ ± 8′	N.G.C. 5128	Continuous. Weak emission lines, $H\beta$, $H\gamma$, $H\delta$, and λ 4686	Unresolved nebula crossed by a marked obscuring band

determined and details of these objects are given in the accompanying table.

The position of the source Taurus A agrees, within the limits of experimental error, with that of the Crab nebula. This is believed to be the expanding shell of the supernova of A.D. 1054. The present dimensions of the nebula are 4′ × 6′, and photographs taken with different wave-length ranges show considerable fine structure, including $H\alpha$ filaments. The emission from the nebulosity is some 600 times that of the central star identified by Baade[2]. Minkowski[3] considers that most of the radiation from the central star is in the far ultra-violet and estimates the surface temperature as 500,000° K. He suggests a temperature of 50,000° K. for the nebulosity and explains a low ratio of $H\alpha$ to [N II] emission as being due to a deficiency of hydrogen. It should be pointed out, however, that a much higher temperature would have a similar effect on this ratio and at the same time increase the electron density. Intensity measurements we have made at 100 Mc./s., assuming a source size of 5′, give an apparent temperature of 2,000,000° K. for Taurus A. The present estimates of density and temperature in the Crab nebula would fall well short of explaining this result by strictly thermal processes. Non-thermal components resulting from the expansion of the nebula do not, however, seem unlikely.

Both N.G.C. 5128 and N.G.C. 4486 are generally classed as extra-galactic nebulæ, though Paraskevoulos[4] considers that the position of the obscuring band splitting the disk of N.G.C. 5128 into halves is unusual. Such obscuring bands are normally seen in nebulæ viewed on edge. Neither of these objects has been resolved into stars, so there is little definite evidence to decide whether they are true extra-galactic nebulæ or diffuse nebulosities within our own galaxy. If the identification of these objects with the discrete sources of radio-frequency energy can be accepted, it would tend to favour the latter alternative, for the possibility of an unusual object in our own galaxy seems greater than a large accumulation of such objects at a great distance.

Full details of the technique and observations involved in the determination of the positions of these three discrete sources will be published elsewhere[5].

This work is part of the research programme of the Division of Radiophysics, Commonwealth Scientific and Industrial Research Organisation, Australia.

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[1] Bolton, J. G., *Nature*, **162**, 141 (1948).
[2] Baade, W., *Astrophys. J.*, **96**, 188 (1942).
[3] Minkowski, R., *Astrophys. J.*, **96**, 199 (1942).
[4] Paraskevoulos, J. S., *Harvard Bull.* 890, 1 (1935).
[5] Bolton, J. G., and Stanley, G. J., *Aust. J. Sci. Res.*, A (in the press).

Resonance Effects in the Reaction $H_1^3(d,n)He_2^4$

The excitation function of the reaction

$$H_1^3 + H_1^2 = He_2^4 + n + 17{\cdot}6 \text{ MeV}.$$

has been studied over an energy-range of 90–190 keV. using the low-voltage accelerator of this Establishment. A thick target of heavy ice was bombarded with a beam containing tritium ions, and the alpha-particles emitted were observed in a proportional counter. Evidence has been found for a resonance in the cross-section of the reaction for an energy of the incident tritium ions of (155 ± 15) keV. The cross-section at the peak of the resonance is several barns. The experiment was performed as follows.

Hydrogen containing between 0·01 and 0·02 per cent of tritium was admitted to the ion source, and a collimated beam of about 10 μamp. (containing more than 90 per cent monatomic ions) was allowed to fall directly on to a layer of heavy ice held on a cooled silver plate. The amount of charge arriving at the target was measured by observing the rise of potential across a condenser connected between it and earth, care being taken to prevent the emission of secondary electrons or collection of stray electrons by the target. The alpha-particles emitted in a direction at 90° to the incident beam were observed with a proportional counter. The excitation curve of the reaction was obtained by counting the number of alpha-particles entering the counter during the collection of a definite amount of charge by the target, while bombarding with ions of the desired energy. Frequent checks were made of the disintegration yield at a standard energy (usually 144 keV.) to detect any possible changes in the beam composition or in the target efficiency. The results are finally expressed as the ratio of the counts at any desired energy to the counts at the standard energy.

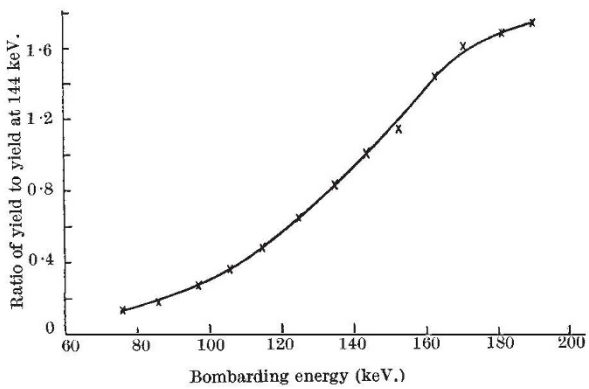


Fig. 1

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